Finance 513 Spring 2026
Advanced Financial Derivatives Martin Widdicks

Problem Set 5
Total 15 points
Due Friday, April 17th at 11:59pm

Instructions. This is an individual assignment. Handwritten solutions are acceptable here, please either scan them to submit.. Of course, you can also typeset your solutions and upload them electronically.

(a) The file name should be: PSn.FAMILYNAME1Initial1.xls, for example my solution to Problem set 1 would have the file name PS1.WIDDICKSM.xls.

1. (3 points) Suppose that the U.S. dollar price of a British pound (GBP) follows the process

$$dS_t = \mu S_t dt + \sigma S_t dX_t$$

where σ, μ are constants. The U.S. dollar and Pound interest rates are constants r and r_P , respectively. Thus, the prices of dollar and euro bonds (or equivalently, money-market accounts) B and B_P follow the processes

$$dB_t = rB_t dt$$
$$dB_{Pt} = r_P B_{Pt} dt$$

with initial conditions $B_0 = 1$ and $B_{P0} = 1$, respectively.

The U.S. dollar price of the GBP bond is $Y_t = S_t B_{Pt}$. You wish to price a call option on the exchange rate, and you will select B_t as the numeraire asset.

(a) (1.5 point) By considering Y_t/B_t or otherwise, determine the dynamics of the U.S. dollar price of a GBP bond, Y_t, under the risk-neutral probability, i.e. what is dY_t?

Using Ito's Lemma

$$\boldsymbol{dY_t = (\mu + r_P)Y_t dt + \sigma Y_t dX_t}$$

And then

$$\boldsymbol{d\left(\frac{Y_t}{B_t}\right) = \left(\mu + r_p - r\right)\frac{Y_t}{B_t}dt + \sigma\frac{Y_t}{B_t}dX_t}$$

As Y/B is a martingale under Q then via the Girsanov theorem $dX^Q = dX - (\mu + r_P - r)/\sigma$ and so

$$\boldsymbol{dY_t = rY_t dt + \sigma Y_t dX_t}$$

(b) (1 point) Using your answer to a) determine the dynamics of the U.S. dollar price of a British Pound, S_t, under the risk-neutral probability, i.e what is dS_t

***Solution*: In the real world measure**

$$dS_t = \mu S_t dt + \sigma S_t dX_t$$

As $dX^Q = dX - (\mu + r_P - r)/\sigma$ and so

$$dS_t = (r - r_P)S_t dt + \sigma S_t dX^Q_t.$$

(c) (1 point) Using your answer to b) and the probabilistic solution, obtain an explicit formula giving the value of a European call option giving the owner the right to buy one GBP for an exercise price of K dollars on date T.

Solution.

Then, by our rules of thumb

$$C(S,0) = e^{-r_P T} S_0 N(d_1) - e^{-rT} K N(d_2)$$

With

$$d_1 = \frac{ln\left(\frac{S_0}{K}\right) + \left(r - r_p + 1/2\sigma^2\right)T}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T}$$

2. (3 points) Assume that we are in the Black-Scholes world where

$$dS_t = \mu S_t dt + \sigma S_t dX_t,$$
$$dB_t = rB_t dt$$

and choose the **stock as the numeraire asset**.

(a) (1/2 point) If the stock is the numeraire asset, what is true about the process followed by the ratio of the stock price to the bond price, B_t/S_t?

It must be a martingale under the appropriate measure.

(b) (1 point) By using your answer to part (a), Ito's lemma, and Girsanov's theorem, determine the stochastic process followed by S_t when S_t is the numeraire (this is **not** as trivial as it sounds).

Let Y = B/S, We have $\partial Y/\partial t = 0$, $\partial Y/\partial S = -B/S^2$, $\partial^2 Y/\partial S^2 = -2B/S^3$, $\partial Y/\partial B = 1/S$. From Itô's formula under the real world probability,

$$dY_t = (r - \mu + \sigma^2)Y_t dt - \sigma Y_t dX_t$$

So, introduce a new probability measure, Q, such that $X = \widetilde{X} - \gamma t$ is a Brownian motion. Then

$$dY_t = (r - \mu + \sigma^2)Y_t dt - \sigma Y_t\left(d\widetilde{X_t} - \gamma dt\right) = \sigma Y_t d\widetilde{X_t}$$

if $\gamma = -(r-\mu+\sigma^2)/\sigma$ in this case, then

$$dS_t = (r+\sigma^2)S_t dt + \sigma S_t d\widetilde{X}$$

(c) (1/2 point) Now consider a European call option, V_t. Write down the value of the option in terms of an expectation when S_t is the numeraire.

According to the probabilistic solution, then under the measure associated with S_t as the numeraire asset, V/S is a martingale and so $V_0 = S_0 E^Q_0[V_T/S_T]$.

(d) (1 points) Using your answers to parts (b) and (c) write V_t as the sum of two integrals and **solve the integral with S in it** (this is the $N(d_1)$ one! Leave the other integral as it is more difficult with this numeraire!). Was this easier than evaluating this term (i.e $N(d_1)$) when B_t was numeraire?

Now we have that

$$V_0 = S_0 E_0^Q\left[\max\left(1-\frac{K}{S_T},0\right)\right] = \frac{S_0}{\sqrt{2\pi}}\int_L^\infty e^{-x^2/2}dx - \frac{K}{\sqrt{2\pi}}\int_L^\infty e^{-x^2/2}e^{-\left(r-\frac{1}{2}\sigma^2\right)T-\sigma\sqrt{T}x}dx$$

Where

$$L = \frac{\ln\left(\frac{K}{S_0}\right)-\left(r+\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}},$$

And so the first term is simply $S_0N(d_1)$ with the usual definition of d_1. This is very straightforward!

Question 3 (3 points): A practical example of using Ito's Lemma

Suppose that the prices of two (non-dividend paying) stocks and a risk-free asset follow the processes:

$$dS_{1t} = \mu_1 S_{1t}dt + \sigma_1 S_{1t}dX_{1t}$$
$$dS_{2t} = \mu_2 S_2 dt + \sigma_2 S_{2t}dX_{2t}$$
$$dB_t = rB_t dt$$

where X_{1t} and X_{2t} are correlated Brownian motions with correlation ρ. Additionally, μ_1, μ_2, σ_1, σ_2, r are positive constants.

a) Consider the process for, which is the product of the two stocks, $Y_t = S_{1t}S_{2t}$. Find dY_t. Show that it is possible to write dY_t in terms of a new Brownian motion X_{3t}.

[YOU ANSWERED THIS ON PS4]

Now, consider a geometric average call option where the payoff is $V(S_{1T}, S_{2T}, T) = \max((S_{1T}S_{2T})^{1/2} - K, 0)$

b) Write down this payoff in terms of Y_T.

[YOU ANSWERED THIS ON PS4]

c) Determine the stochastic process followed by $Z_t = Y_t^{1/2}$.

[YOU ANSWERED THIS ON PS4]

Note you can always combine two Brownian motions to create a new one. In particular, we can write $\sigma_3 dX_{3t} = a dX_{1t} + b dX_{2t}$ where $\sigma_3 = \sqrt{(a^2+b^2+2\rho ab)}$, and you can check that $E[dX_{3t}] = 0$ and $Var[dX_{3t}] = dt$.

d) Using your answer to c) determine the real world expected return of the geometric average of two stocks?
[YOU ANSWERED THIS ON PS4]

We are now going to value the Option by using the risk-free asset, B_t, as the numeraire asset

e) Define $\eta_t = Z_t/B_t$. What are the dynamics of η under the original probability, ***P***? (That is, what is $d\eta$**?** Again, I would write your answer in terms of a single Brownian motion X_3).

(1 point)

Assuming that Z_t is a traded asset

$$d\eta_t = (\tfrac{1}{2}(\mu_3 - \tfrac{1}{4}\sigma_3^2) - r)\eta_t\, dt + \tfrac{1}{2}\sigma_3\, \eta_t\, dX_3$$

f) By the fundamental theorem of finance what must be true about η_t under the equivalent martingale measure ***Q***?

Under Q, η_t must be a martingale and so $d\eta_t = \tfrac{1}{2}\sigma_3\, \eta_t\, d\widetilde{X}_3$

Where $dX_3 = d\widetilde{X}_3 - \gamma dt$ where $\gamma = (\tfrac{1}{2}(\mu_3 - \tfrac{1}{4}\sigma_3^2) - r)/\tfrac{1}{2}\sigma_3$

(0.5 points)

g) By applying Giranov's theorem and your answers to c), e) and f) determine the process for Z_t under the measure ***Q***.

And so $dZ_t = rZ_t dt + \tfrac{1}{2}\sigma_3\, Z_t\, d\widetilde{X}_3$

h) By analogy to the Black-Scholes equation, or otherwise, write down the USD value of the option $V(S_1, S_2, 0)$.

(1 points)

By analogy to BSM we have $V(Z_0, 0) = e^{-rT}(Z_0 e^{rT} N(d_1) - KN(d_2))$

Where $$d_1 = \frac{\ln\left(\frac{Z_0}{K}\right) + \left(r + \frac{1}{4}\sigma_3^2\right)T}{\frac{1}{2}\sigma_3\sqrt{T}},\quad d_2 = d_1 - \frac{1}{2}\sigma_3\sqrt{T}$$

And upon simplifying

$$V(S_1, S_2, 0) = e^{-rT}\left(\sqrt{S_1 S_2}\, e^{rT} N(d_1) - KN(d_2)\right)$$

Where $$d_1 = \frac{\ln\left(\frac{\sqrt{S_1 S_2}}{K}\right) + \left(r + \frac{1}{4}\sigma_3^2\right)T}{\frac{1}{2}\sigma_3\sqrt{T}},\quad d_2 = d_1 - \frac{1}{2}\sigma_3\sqrt{T}$$

And $\sigma_3 = \sqrt{(\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2)}$

Question 4 (6 points): Final exam practice

Let B_t be a USD-money market account, B_{Yt} be a Japanese Yen denominated money market account and S_t be the price of the Nikkei Index denominated in Japanese Yen (JPY), the exchange rate (or the value of JPY in USD) is e_t. These prices and rates follow the following stochastic processes:

$$dB_t = rB_t dt$$
$$dB_{Yt} = r_Y B_{Yt} dt$$

$$de_t = \mu_e e_t dt + \sigma_e e_t dX_{1t}$$
$$dS_t = (\mu - d)S_t dt + \sigma S_t dX_{2t}$$

$$E[dX_{1t} dX_{2t}] = \rho dt$$

Where X_{1t} and X_{2t} are Brownian motions under the $\boldsymbol{P}$ measure and where the correlation, ρ, the risk-free rates, r and r_Y, the drifts, μ, μ_e, dividend yield, d, and standard deviation, σ and σ_e, terms are constants. We wish to value derivative products with maturity T whose payoff depends upon the JPY price of the Nikkei but the payoff is in USD (this type of product is called a quanto.)

a) First determine the stochastic process followed by the dollar value of the JPY money market account, that is $B_{YDt} = e_t B_{Yt}$

$$\begin{aligned} \boldsymbol{dB_{YDT}} &= \boldsymbol{e_t dB_{Yt} + B_{Yt} de_t} \\ &= \boldsymbol{r_Y B_{YDt} dt + (\mu_e B_{YDt} dt + \sigma_e B_{YDt} dX_{1t})} \\ &= \boldsymbol{(r_Y + \mu_e) B_{YDt} dt + \sigma_e B_{YDt} dX_{1t}} \end{aligned}$$

b) Unfortunately, as the Nikkei pays a dividend, we must consider the 'Dividend Asset', $D_t = S_t e^{-d(T-t)}$ (In JPY). By using Ito's lemma (or otherwise) determine the stochastic process followed by D_t

$$\boldsymbol{dD_t = \mu D_t dt + \sigma D_t dX_{2t}}$$

c) Next, determine the stochastic process followed by the dollar value of the dividend asset $U_t = e_t D_t$ (Note: do not combine dX_1 and dX_2 here)

$$\boldsymbol{dU_t = (\mu + \mu_e + \rho\sigma\sigma_e) U_t dt + \sigma U_t dX_{2t} + \sigma_e U_t dX_{1t}}$$

d) Define $Y_t = \frac{B_{YDt}}{B_t}$ and $Z_t = \frac{U_t}{B_t}$. Determine the stochastic processes followed by Y_t and Z_t under the original probability, $\boldsymbol{P}$?

$$\boldsymbol{dY_t = (r_Y + \mu_e - r) Y_t dt + \sigma_e Y_t dX_{1t}}$$

$$\boldsymbol{dZ_t = (\mu + \mu_e + \rho\sigma\sigma_e - r) Z_t dt + \sigma Z_t dX_{2t} + \sigma_e Z_t dX_{1t}}$$

<u>For ease of valuation, we will now change measure to *Q* where B_t is the numeraire asset.</u>

e) By the fundamental theorem of finance what must be true about Y_t and Z_t under measure $\boldsymbol{Q}$?

They are both martingales

f) **<u>Now consider the stochastic process followed by Y_t</u>** (and your answer to e). By applying Girsanov's theorem, for the new probability measure, ***Q***, find the new Brownian motion $\tilde{X}_{1t}$ (under ***Q***). Please note the value of γ_1 for transforming X_{1t} into $\tilde{X}_{1t}$.

Choose $\boldsymbol{\gamma_1 = \frac{r_Y + \mu_e - r}{\sigma_e}}$ **and then**

$$dY_t = Y_t d\widetilde{X_{1t}}$$

g) **<u>Now consider the stochastic process followed by Z_t</u>** (and your answer to e). By applying Giranov's theorem a second time, find the new Brownian motion $\tilde{X}_{2t}$ (under ***Q***) in the new measure. (***You will need your results from part d) and part f)***). Please note the value of γ_2 for transforming X_{2t} into $\tilde{X}_{2t}$.

Now, $\boldsymbol{dX_{1t} = d\widetilde{X_{1t}} - \gamma_1 dt}$ **and so**

$$dZ_t = (\mu + \mu_e + \rho\sigma\sigma_e - r)Z_t dt + \sigma Z_t dX_{2t} + \sigma_e Z_t\left(d\widetilde{X_{1t}} - \gamma_1 dt\right)$$
$$= (\mu + \rho\sigma\sigma_e - r_Y)Z_t dt + \sigma Z_t(d\widetilde{X_{2t}} - \gamma_2 dt) + \sigma_e Z_t \widetilde{dX_{1t}}$$

Choosing $\boldsymbol{\gamma_2 = \frac{(\mu + \rho\sigma\sigma_e - r_Y)}{\sigma}}$ **and then**

$$dZ_t = \sigma Z_t d\widetilde{X_{2t}} + \sigma_e Z_t \widetilde{dX_{1t}}$$

h) Now, using results from parts b), f) and g) show that the stochastic process for S_t under ***Q*** is

$$dS_t = (r_Y - d - \rho\sigma\sigma_e)S_t dt + \sigma S_t d\tilde{X}_{2t}$$

Now we have that,

$$dS_t = (\mu - d)S_t dt + \sigma S_t\left(d\widetilde{X_{2t}} - \gamma_2 dt\right)$$
$$= (\mu - d - \mu - \rho\sigma\sigma_e + r_Y)\, S_t dt + \sigma S_t d\widetilde{X_{2t}}$$
$$= (r_Y - d - \rho\sigma\sigma_e)\, S_t dt + \sigma S_t d\widetilde{X_{2t}}$$

Finally, consider a Quanto American put option with payoff (in USD) of $V_T = \max(K - S_T, 0)$ that may be exercised at any time to receive $K - S_t$, where S_t is the value of the Nikkei (In JPY) at time t. The current date is $t = 0$.

i) By using the results from h) above, provide details of a CRR binomial tree algorithm (i.e. what are u, d, q, $V_{N,j}$, $V_{i,j}$) to value the American put option.

$$u = e^{\sigma\sqrt{\Delta t}}, d = e^{-\sigma\sqrt{\Delta t}}, q = \frac{e^{(r_Y - d - \rho\sigma\sigma_e)\Delta t} - d}{u - d}$$

$$V_{N,j} = \max\left(K - S_{N,j}, 0\right)$$
$$V_{i,j} = \max\left(K - S_{i,j}, e^{-r\Delta t}(qV_{i+1,j+1} + (1-q)V_{i+1,j+1}\right)$$
$$S_{i,j} = S_0 u^i d^{j-i}$$